

BUILDING THE HUMAN CIRCULATORY SYSTEM MODEL



FEEL YOUR HEARTBEAT!



Can you feel your pulse? Try
your wrist or neck! 💖

Your heart pumps blood like a
water motor — non-stop!

MEET RAJU RED-CELL!

 Heart pumps blood through blood vessels — just like Indian Railways connects every city across the country!

 Red blood = oxygenated (full of oxygen)
 Blue blood = deoxygenated (oxygen used up)
Spot the difference!

 Blood's journey: Body → Heart → Lungs (like a CNG station, refuelling with oxygen!) → Heart → Body. Round and round!

 Valves act like Metro turnstiles — they only let blood move ONE way, preventing it from flowing backwards!

Raju Red-Cell travels your whole body in less than a minute — amazing!

LET'S GET BUILDING!



Materials You Need
2 syringes, clear tubing, check valves, red & blue food colouring, cardboard template, zip ties or tape, small cups

Step 1 — Connect It Up
Attach the clear tubing and check valves to both syringes. Make sure the valves face the correct direction so water flows one way only!

Step 2 — Add Colour
Fill one syringe with red water (oxygenated blood) and the other with blue water (deoxygenated blood). Watch the colours move through the tubing!

Step 3 — Mount & Display
Secure your tubing and syringes onto the cardboard silhouette using zip ties or tape. Your circulatory system model is ready to pump!

Top Tip: Ask a partner to hold the cardboard while you attach the tubing!



Watch the Blood Flow!

Push the syringe — see how
valves keep blood moving forward,
just like a turnstile!

PULSE RECHECK!

Jump 30 secs — feel
the difference!



QUICK QUIZ — WHAT DID WE LEARN?

Q1: What colour shows oxygenated blood?

Q2: What prevents blood from flowing backward?

Q3: Why does your heart beat faster when you run?



Let's see what you remember! Think carefully and answer each question below.





GREAT JOB,
SCIENTISTS!

